Markscheme

Mathematics: analysis and approaches
Practice question bank

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7. (a)

Consider the expression $\dfrac{(x^3y^{-2})^4(x^{-1}y^5)^2}{(xy)^{-3}}$.

Simplify it to the form x^ay^b, and find a and b.

$$(x^3y^{-2})^4 = x^{12}y^{-8}$$ **A1**

$$(x^{-1}y^5)^2 = x^{-2}y^{10}$$ **A1**

multiply the numerator terms: $x^{12-2}y^{-8+10} = x^{10}y^2$ *(M1)*

divide by $(xy)^{-3} = x^{-3}y^{-3}$ *(M1)*

$$a = 13, \ \ b = 5$$ **A1**

[5 marks]

(b)

Hence simplify $\sqrt{x^ay^b}$ (using your values of a and b) to the form x^py^q, and find p and q.

$$\sqrt{x^{13}y^5} = x^{\frac{13}{2}}y^{\frac{5}{2}}$$ *(M1)*

$$p = 6.5, \ \ q = 2.5$$ **A1**

[2 marks]